Analog Computer

User Manual

---

1. Introduction

The **Analog Computer** is designed to perform real-time mathematical operations on two analog input signals. The system supports **addition**, **subtraction**, **multiplication, differentiation** and **integration** using a dual-interface input architecture and provides a continuous analog output.

---

2. Functional Description

The **Analog Computer** processes two independent analog input signals (**Input A** and **Input B**). Based on the selected operation mode, the unit computes the result instantaneously and delivers it as an analog output signal.

Supported Operations:

- Addition
- Subtraction
- Multiplication
- Differentiation
- Integration

---

3. Technical Specifications

Parameter	Specification
Input Interfaces	Dual analog inputs (A & B) for each operation
Input Range	±10 V (typical)
Output Range	±10 V (operation dependent)
Computation Type	Continuous-time analog
Multiplication Scaling	Adjustable gain and input gain can be selected
Power Supply	±12 V / ±15 V DC (model dependent)

---

4. Panel Description

Operations can be done independently. Addition, subtraction and multiplication can be done parallelly by providing each input required signals.

- **Input A Terminal:** Primary analog input
- **Input B Terminal:** Secondary analog input

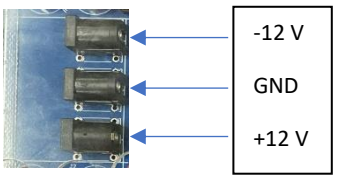
- **Output Terminal:** Computation result

---

5. Operating Procedure

1. Verify correct power supply connectionsapply ±12 V supply to the DC input ports as shown in the figure.
2. Switch ON the Analog Computer.
3. Connect the analog signals to **Input A** and **Input B**.
4. Set the gain according to a known reference operation and fix the gain
5. For subtraction operation feed the minuend to the **Input A** and the subtrahend to the **Input B**.
6. Observe the output signal using a voltmeter or oscilloscope.

Note: Ensure all input signals remain within the specified voltage limits.

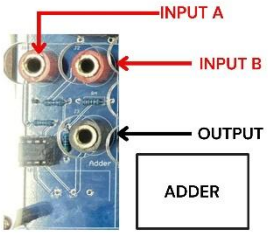


---

6. Mathematical Operations

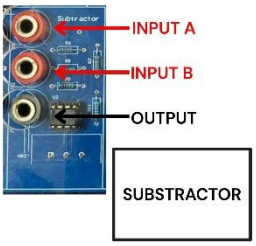
Addition Mode

```
Vout = K(aVA + bVB)

(0 < K < 2, a = [2 or 3], b = [2 or 3])
```



Subtraction Mode

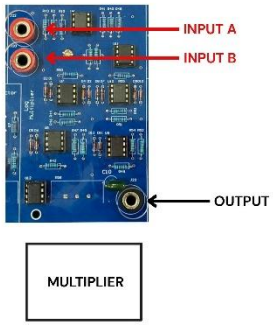
```
Vout = K(aVA - bVB)

(0 < K < 2, a = [2 or 3], b = [2 or 3])
```



Multiplication Mode

```
Vout = k × VA × VB
```

where k is a variable scaling constant.



- To select a and b constant values , look in the top upper section and feed the inputs to A1, A2,B1 or B2, A1 and B1 will give outputs from A=2 and B =2 output ports, A2 and B2 will give outputs from A=3 and B =3 output ports. These outputs corresponds to the a and b constant values

7. Safety Precautions

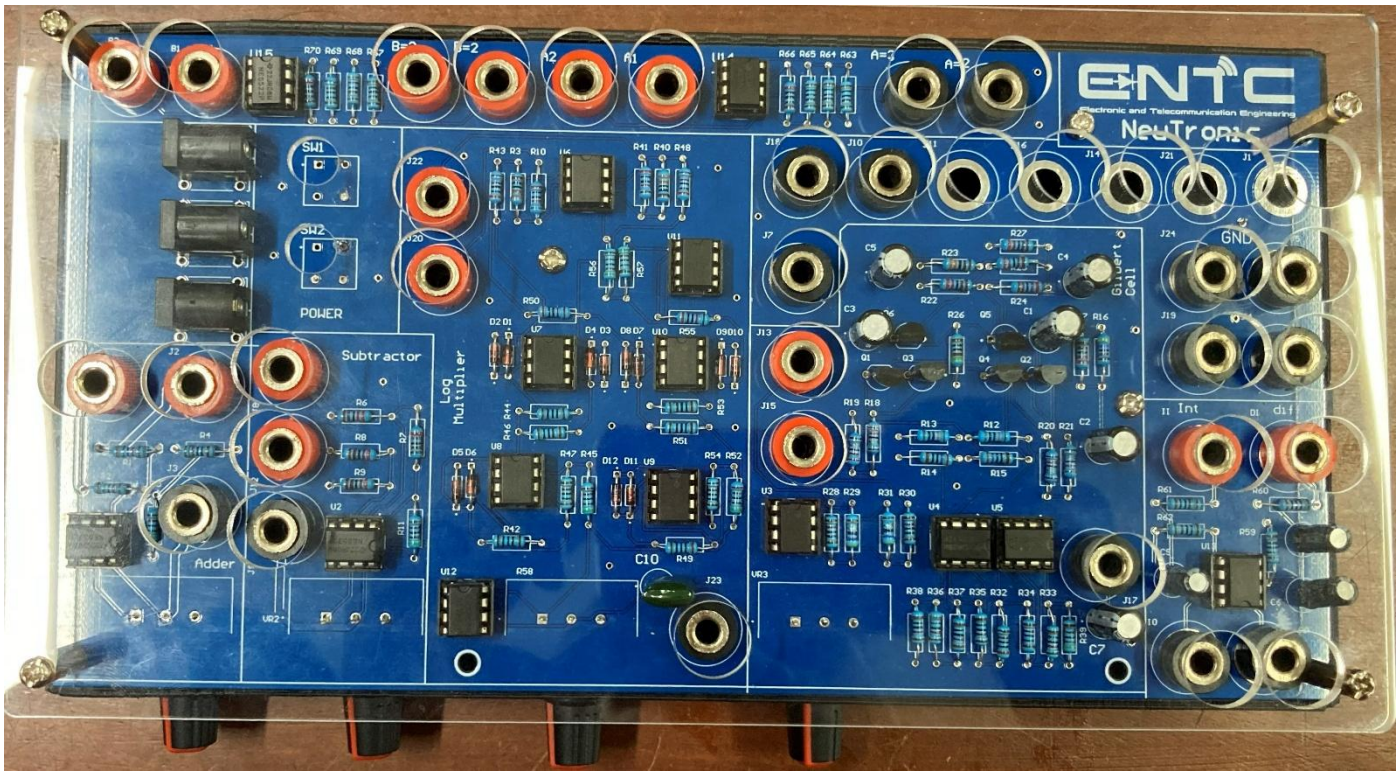
- Do not exceed maximum input voltage ratings
- Disconnect power before modifying connections
- Avoid moisture and excessive heat exposure
- Ensure proper grounding for accurate operation

8. Applications

- Analog signal processing
- Control systems analysis
- Educational laboratory experiments
- Real-time mathematical modeling
- Telecommunications

9. Maintenance

The ACU requires minimal maintenance. Periodic calibration is recommended to maintain accuracy. Keep dust free and controlled room temperature



End of document.